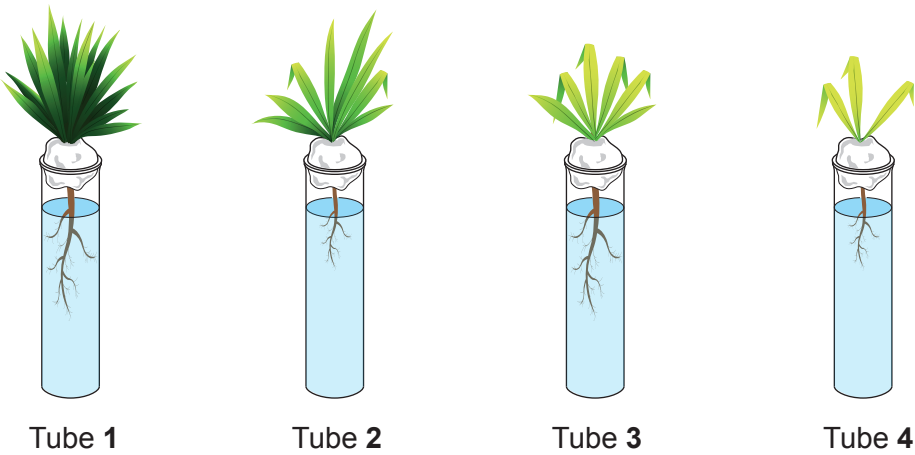


7. Julius von Sachs was a 19th century German scientist who studied the effect of plant nutrients on growth. He developed a growth solution which contained all the nutrients a plant needed for healthy growth.

A version of one of his experiments is shown below.

- 1. Seedlings of the same age were placed on cotton wool in four test tubes, each containing an equal volume of a different nutrient solution.
- 2. All four test tubes were placed on a shelf in full sunlight.
- 3. The growth of the seedlings was observed and recorded after one month (**Image 7.1**).

Image 7.1



	Tube 1	Tube 2	Tube 3	Tube 4
Content of growth solution	all nutrients present	no nutrient X	no nutrient Y	no nutrient Z
Description of growth after one month	healthy growth	poor root growth	yellowing of leaves	poor growth

(a) Identify the missing nutrients. [3]

Nutrient X =

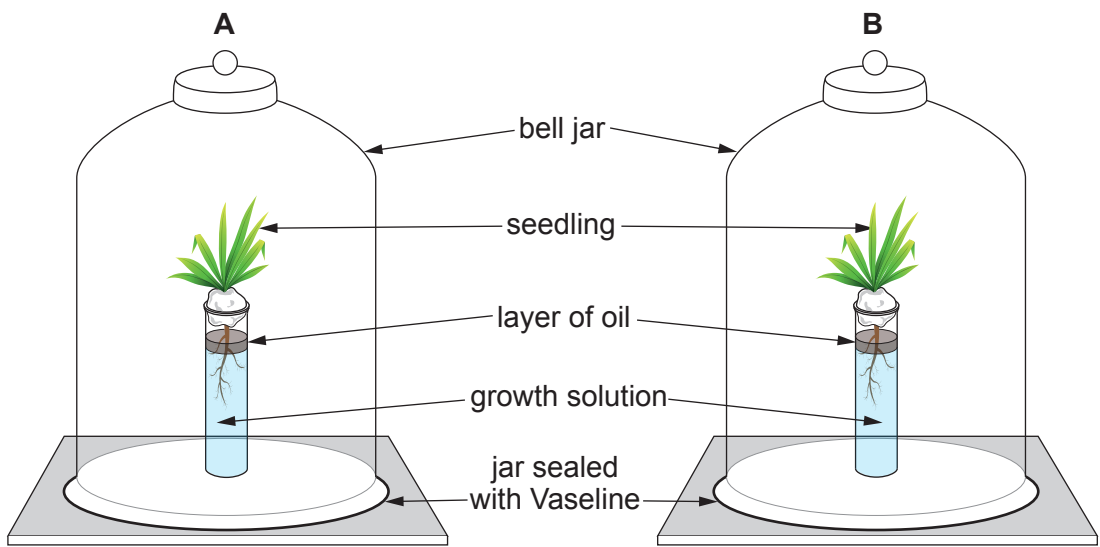
Nutrient Y =

Nutrient Z =



(b) Students modified the experiment by taking two seedlings of similar age and placing them in a sealed bell jar as shown in **Image 7.2**.

Image 7.2



Bell jar A

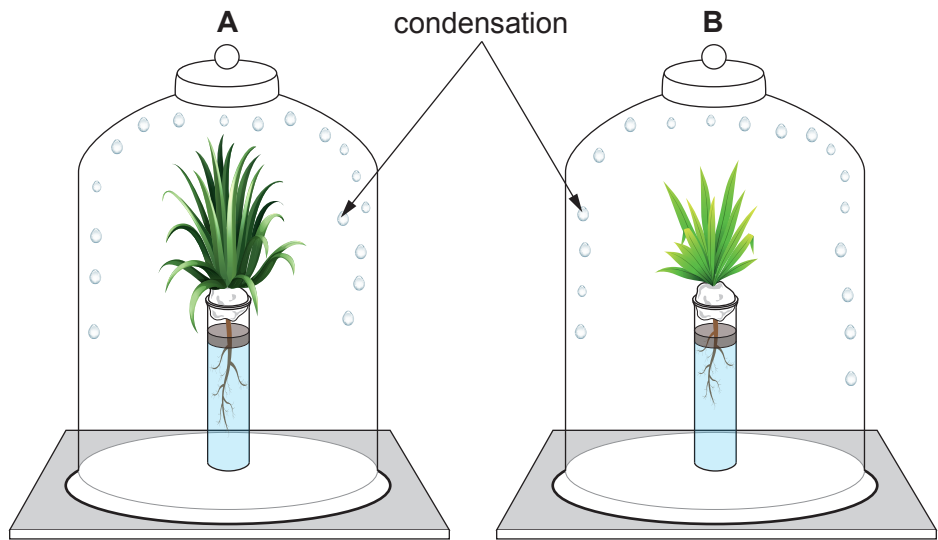
Seedling in growth solution containing all nutrients

Bell jar B

Seedling in growth solution containing all nutrients **and** a chemical which reduces the rate of respiration in root hair cells

Image 7.3 shows the appearance of the apparatus after 2 weeks.

Image 7.3



Examiner
only

(i) Suggest the purpose of the layer of oil in the experiment. [1]

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(ii) Explain the presence of the condensation (water droplets) inside the bell jars in **Image 7.3**. [2]

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(iii) Using your knowledge of how nutrients are absorbed, explain the difference in growth between the plants in bell jars **A** and **B** during the 2 weeks. [2]

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8

